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Performance Improvement of YOLOv13n for Waste Detection Using Contrast-Limited Adaptive Histogram Equalization in a Web-Based System

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1. Introduction

Global solid waste generation reached approximately 2.01 billion tonnes in 2016 and is projected to surge to 3.40 billion tonnes by 2050[1], [2]. In Indonesia, annual waste generation has reached a critical level of approximately 25 million tons[3], with current data indicating that only 66% is managed effectively[3]. Traditional frameworks struggle to keep pace due to a reliance on manual sorting, which is not only inefficient and prone to categorization errors but also exposes workers to potential physical risks and unhygienic materials found in common waste streams[4], [5], [6]. Artificial Intelligence and computer vision have emerged as revolutionary tools for addressing global waste issues[7]. Deep learning models can automatically extract intricate features from visual data for high-precision classification[8]. Within the field of object detection, single-stage detectors such as You Only Look Once (YOLO) perform detection and classification in a single evaluation step[9]. The YOLO family of algorithms has undergone significant evolution from early versions to high-performance architectures like YOLOv12[10]. Architectural innovations such as anchor-free detection and enhanced feature extraction have consistently improved model speed[11].

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Despite significant algorithmic advancements, detecting waste in complex real-world environments remains a major challenge[12]. Environmental factors such as lighting fluctuations can significantly alter the visibility of an object[13]. Poor illumination and shadows often lead to missed detections, particularly for small or overlapping items[14]. In urban surveillance or underwater scenes, low image contrast degrades the recognition accuracy of deep learning models[15]. Preprocessing via image enhancement is a fundamental strategy for addressing these environmental visibility constraints[4]. Standard steps such as resizing and pixel normalization ensure input consistency for neural networks[16]. Contrast-Limited Adaptive Histogram Equalization (CLAHE) offers a robust mechanism for highlighting features in low-contrast visual data[17]. Integrating robust preprocessing methods can provide the visual clarity necessary for high-precision sorting.[8]

This research explores the tools and technologies, ranging from open-source deep learning frameworks to web-based deployment platforms, that facilitate the automated management of waste. Algorithms serve as the core of these artificial intelligence systems, driving decision-making processes and enabling intelligent sorting behavior[7]. This paper aims to provide a performance evaluation of the techniques and challenges in waste detection using the latest YOLOv13n architecture. We delve into the underlying theories of feature extraction, explore the basis of image enhancement, and discuss the importance of local contrast optimization[7]. Additionally, we survey existing methodologies, ranging from traditional machine learning to deep learning models such as YOLOv5 and YOLOv8, and analyze their limitations in low-contrast scenarios [10]. By addressing these challenges and discussing the integration of Contrast-Limited Adaptive Histogram Equalization, the goal is to serve as a resource for researchers seeking to leverage artificial intelligence for smart city infrastructure[18]. We collected the journal papers from reputable digital databases and searched around 50 papers with keywords such as waste detection and YOLO to establish a theoretical baseline[1]. Next, we analyzed these papers to identify the research gap regarding the newest iterations of real-time detectors[1].

A primary innovation of this study is the performance evaluation of the YOLOv13n model, which represents a new iteration beyond the architectures documented in current literature[10]. This research introduces an original integration of Contrast-Limited Adaptive Histogram Equalization to enhance local contrast in low-visibility waste imagery [19]. By utilizing the nano-sized variant of the architecture, the study optimizes the system for high-speed processing on hardware with limited resources[20]. The proposed system provides a scalable web-based interface that supports modern urban waste management frameworks[21]. This integration facilitates the development of automated sorting machines and recycling bins powered by artificial intelligence[11]. Furthermore, the system helps mitigate health risks associated with the manual handling of sharp or contaminated recyclable materials[22]. This breakthrough provides a replicable methodology that can be integrated into diverse smart city infrastructures[23]. The insights gained from this study contribute to the development of more accurate and robust systems capable of managing urban litter effectively[1].

The next section discusses some relevant research in the field of waste detection using deep learning approaches. Section 3 discusses the methodology used in automated waste

detection. More specifically, information is given about the datasets used, the algorithms used, as well as the experimental steps performed. The results of the experiments conducted in waste detection using the YOLO algorithm are discussed in Sec. 4. From the experimental results and analysis conducted with the various algorithms, Sec. 5 describes the implications and limitations of the findings. Finally, Sec. 6 shows the conclusions and avenues for future research.

2. Related Work

Major headings should be typeset in boldface with the first letter of important words capitalized.

2.1. *Deep learning in waste management system*

The integration of deep learning into waste management systems has emerged as a significant approach to addressing the growing challenge of global waste accumulation, marking a shift from conventional methods toward intelligent automation. Through computer vision techniques, these systems enable real-time detection, localization, and classification of various waste materials. Such approaches have been widely implemented in automated sorting facilities, waste-sorting robots, and smart bin applications that monitor fill levels and support waste separation at the source[24]. The main advantage of deep learning approaches over manual methods lies in their ability to substantially improve operational speed and efficiency, while reducing reliance on human labor that is prone to error and potential health risks from prolonged exposure to waste materials[25].

However, the practical deployment of waste detection systems remains a challenge. System performance is frequently affected by environmental variability, including substantial variations in illumination, unfavorable weather conditions, complex visual backgrounds, and waste objects that are partially occluded, overlapping, or physically deformed[26]. This indicates that although deep learning has become a crucial component in the modernization of waste management, a gap remains between high accuracy achieved in controlled environments and reliable performance in dynamic real world conditions, which continues to be a major challenge to address[24][27].

2.2. *Waste detection using object detection models*

2.2.1. *Transition from image classification to object detection*

The transition from manual waste sorting to automated intelligent systems has been accelerated by advances in Deep Learning, particularly through the use of Convolutional Neural Networks. Early studies primarily relied on image classification approaches that assigned a single label to an entire image. However, recent literature increasingly adopts object detection frameworks, as they enable not only waste

categorization but also precise localization of multiple objects within complex real world scenes, which is essential for automated sorting and robotic manipulation[28].

2.2.2. *Single stage and two stage detection frameworks*

Within the domain of object detection, research has generally been divided into two-stage and single-stage detectors. Although two-stage approaches are often associated with high detection accuracy, a clear trend in the literature favors single-stage detectors, particularly the You Only Look Once family, for waste management applications. This preference is primarily driven by the need for fast inference and computational efficiency, which are critical for real time deployment in automated sorting systems and resource constrained environments[25], [29].

This preference is primarily driven by the computational efficiency and fast inference capability of single stage models, which are essential for real time deployment in resource constrained systems such as reverse vending machines, autonomous cleaning robots, and smart bins[28]. Existing studies consistently report that although two stage detectors can achieve slightly higher precision in certain scenarios, their higher latency limits their practicality for continuous and high-speed waste processing on conveyor belts or mobile monitoring platforms[30][31].

2.2.3. *Application scenarios in waste detection*

The application of object detection models in waste management has formed several research clusters. A prominent area concentrates on the separation of municipal solid waste and recyclables, with models designed to differentiate between organic and inorganic materials[32][33]. A further line of research emphasizes fine grained material classification, which involves distinguishing between visually similar waste materials and relies on the ability of detection models to capture subtle visual differences[34][35]. Furthermore, object detection has been increasingly adopted for environmental monitoring, including the detection of floating debris in river and marine environments, where background dynamics and illumination variability remain challenging[30].

2.2.4. *Limitations and open challenges*

Despite their extensive use, YOLO based detection models continue to face robustness limitations. Numerous studies report performance degradation when these models operate under complex and unconstrained real-world conditions[26][36]. Standard models frequently struggle with small waste objects, such as cigarette butts or bottle caps, which occupy minimal pixel areas and are easily obscured by cluttered backgrounds[25]. Additionally, challenges related to object occlusion, overlapping

waste items, and variations in lighting conditions continue to reduce the reliability of automated detection systems [36].

These challenges highlight an existing research gap, as most studies focus on architectural modifications, while the potential of image enhancement and preprocessing techniques to address environmental variability and low contrast remains underexplored[28][25]. Consequently, further investigation is required to develop integrated systems that combine lightweight detection models with robust image processing strategies to ensure high accuracy and efficiency in unconstrained waste management scenarios.

2.3. *Performance improvement through image enhancement*

Beyond environmental factors, degraded image quality has been widely recognized as a key contributor to reduced detection performance. Low contrast, uneven illumination, shadows, and motion blur can obscure critical visual cues, making it difficult for detection models to extract discriminative features, particularly in cluttered scenes or low light conditions[37]. To address these challenges, much of the existing literature emphasizes the use of data augmentation strategies during the training phase. These approaches seek to improve model generalization by introducing artificial variability, commonly through geometric transformations and photometric adjustments such as changes in brightness, exposure, and contrast, encouraging models to become more resilient to adverse lighting conditions [24]. In parallel, architectural enhancements such as attention mechanisms and multi scale feature fusion have been investigated to improve feature discrimination in cluttered scenes[26]. While these methods enhance model robustness, they fundamentally address the symptom via training complexity rather than correcting the input quality itself.

In contrast, the use of image enhancement techniques as a preprocessing step remains relatively underexplored within the broader waste detection literature. Although some domain specific studies have applied enhancement or restoration methods, particularly in specialized settings, reported outcomes vary considerably[30]. This variability suggests that the role of image enhancement in improving detection performance has not been systematically examined, especially in relation to lightweight object detection models operating under practical constraints.

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